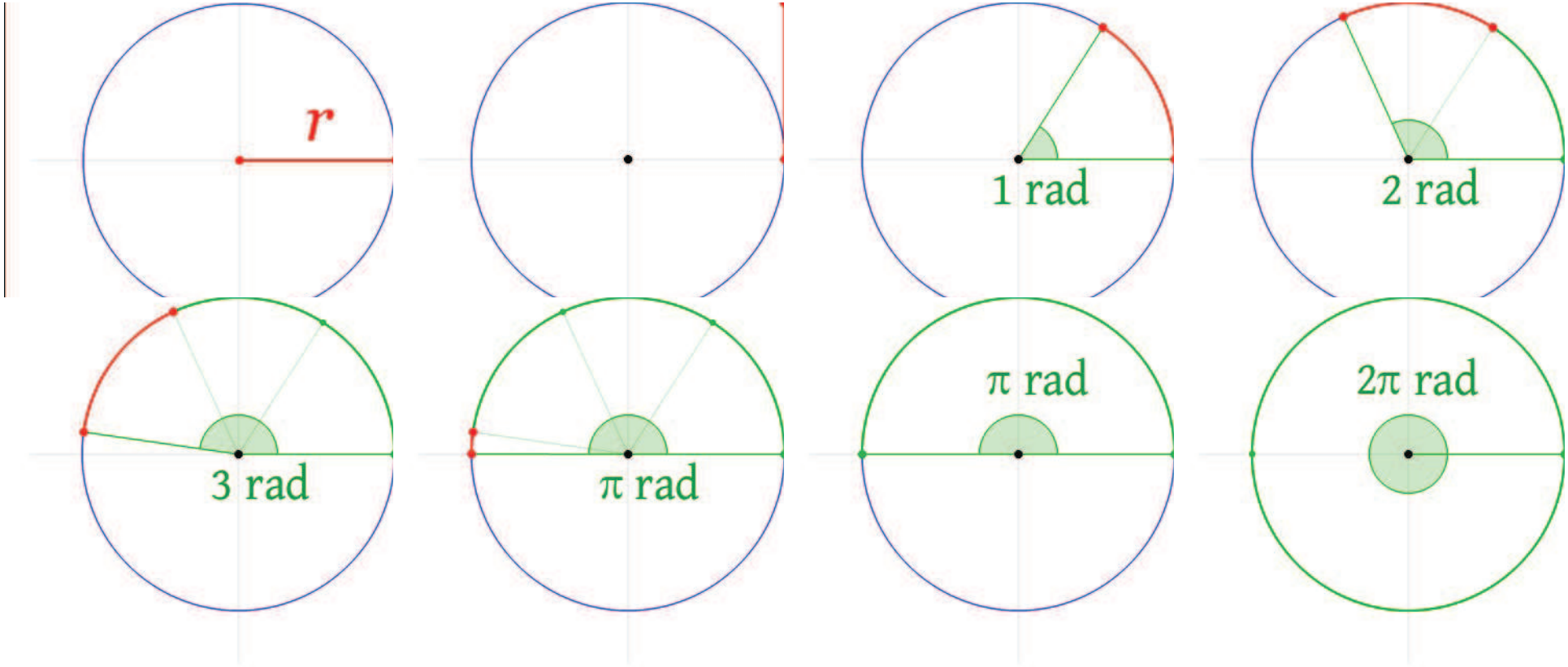
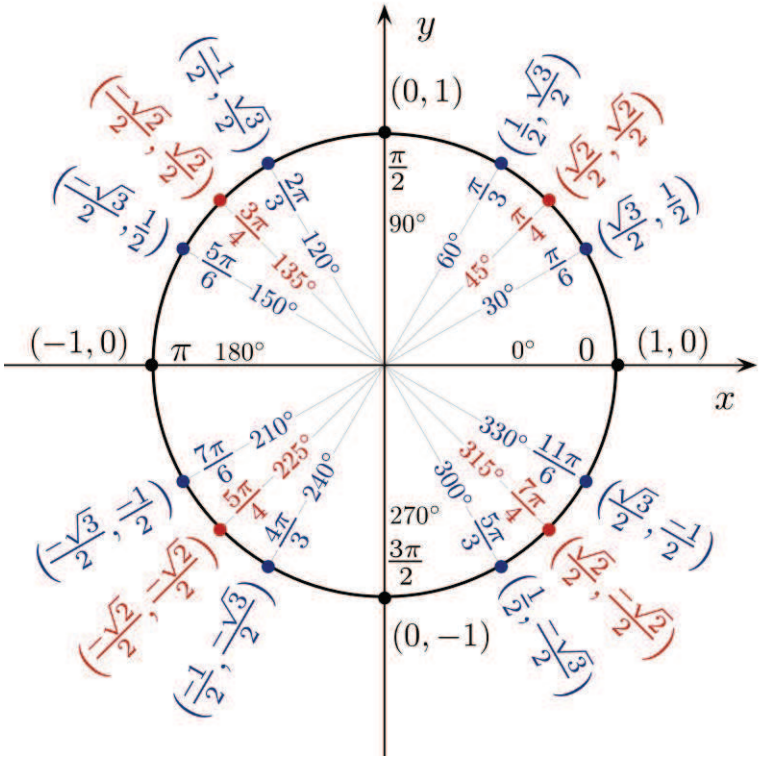
4 Тригонометрични и обратни тригонометрични функции

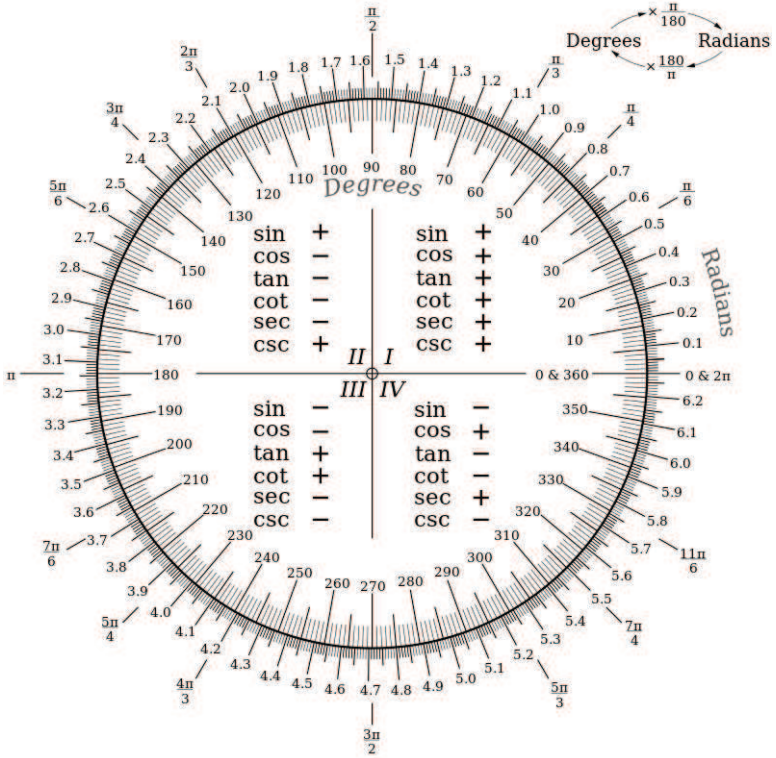
4.1 Тригонометрични функции

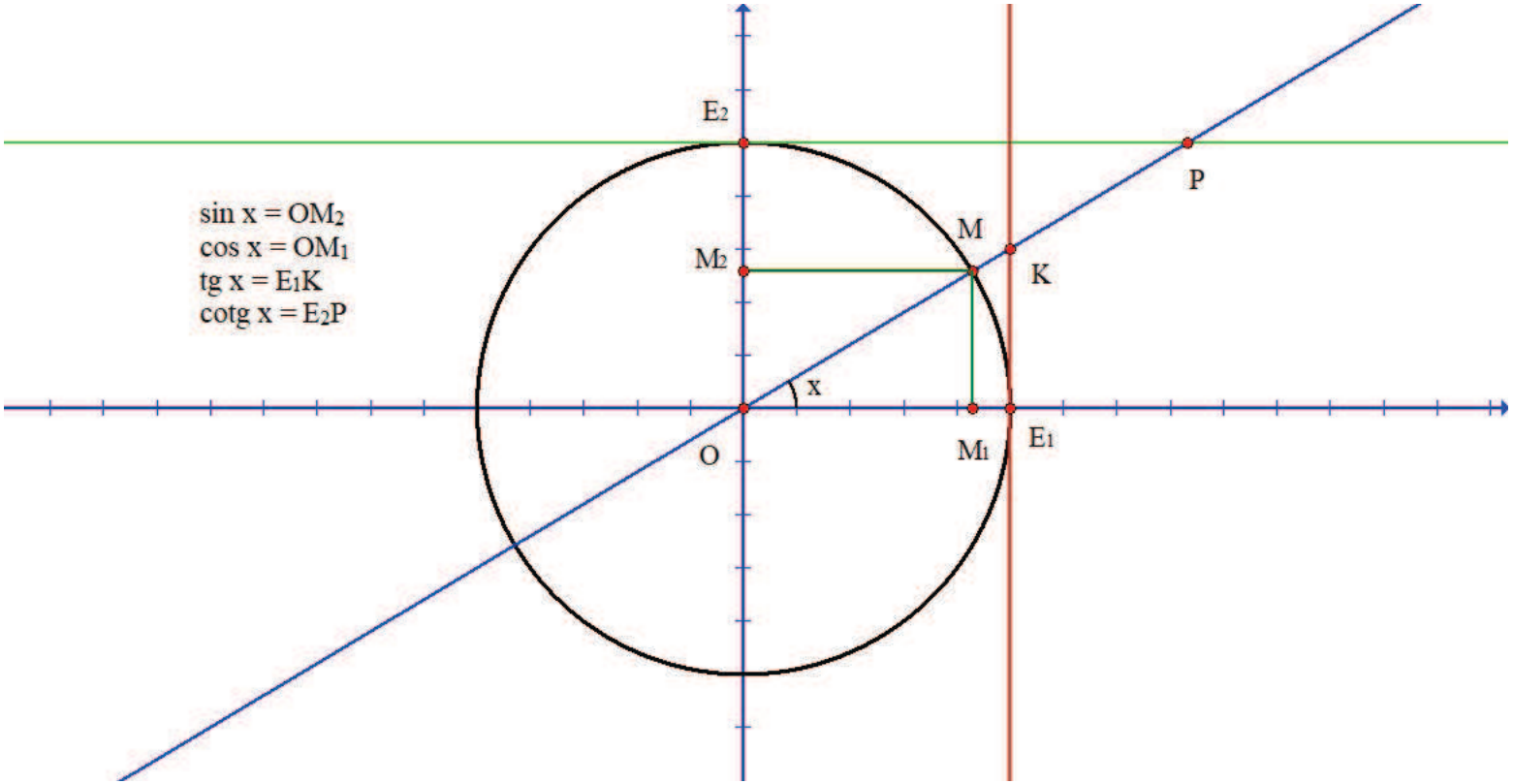
4.1.1 Радиан



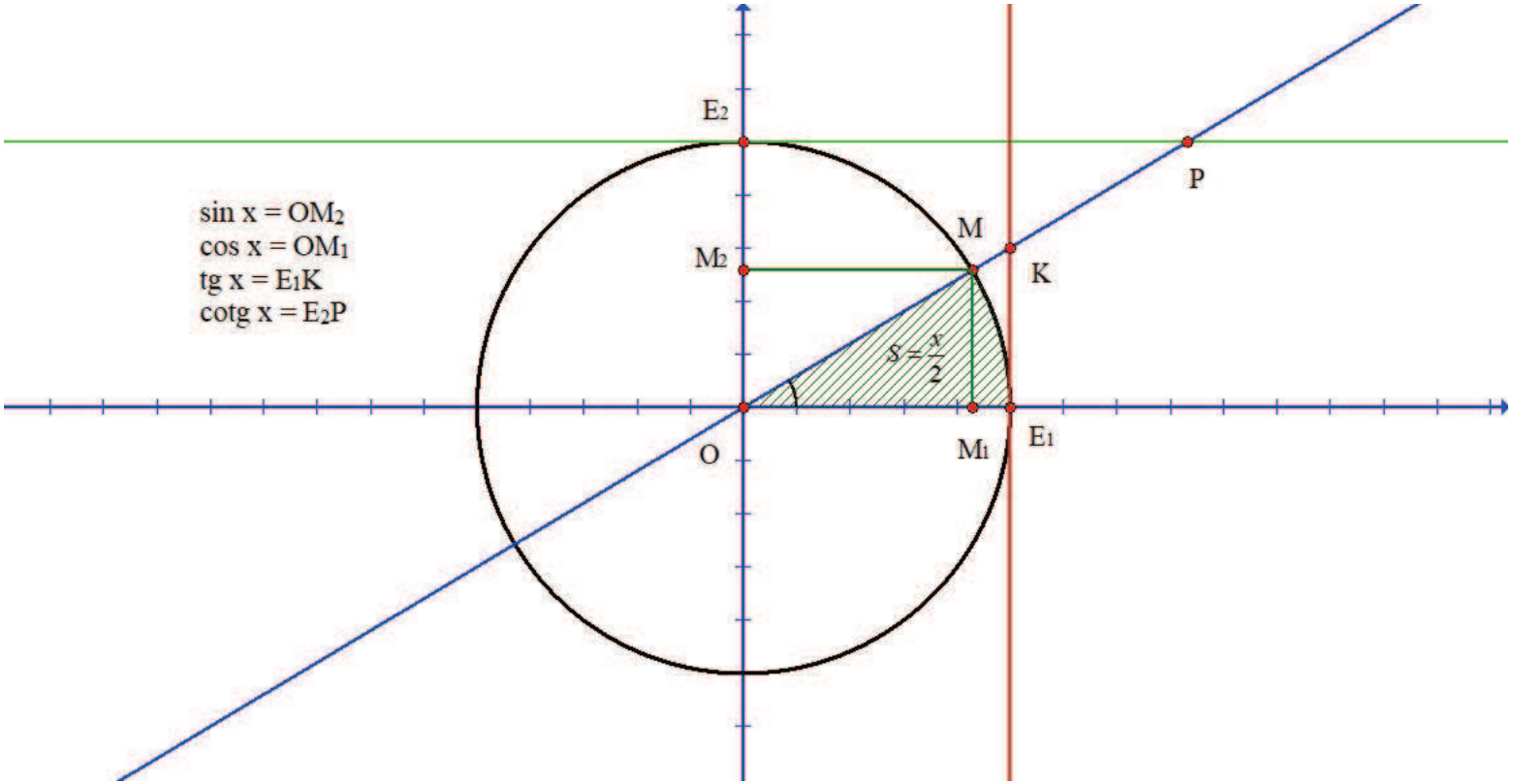
4.1.2 Тригонометрична окръжност






sin x = OM2
cos x = OM1
tg x = E1K
cotg x = E2P
E2
P
M
M2
K
x
O
M1
E1

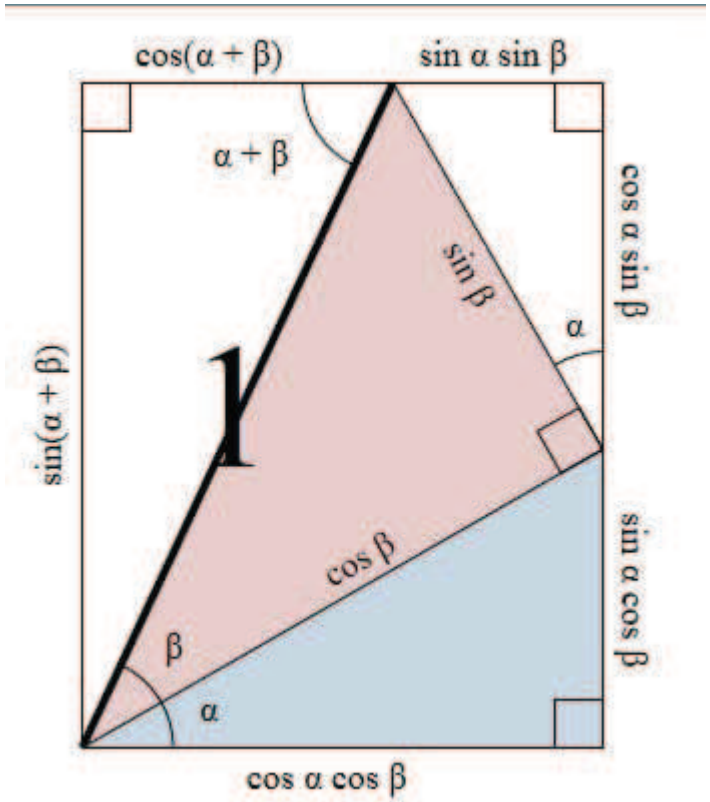
4.1.3 Аргументът като лице



4.1.4 Основни равенства

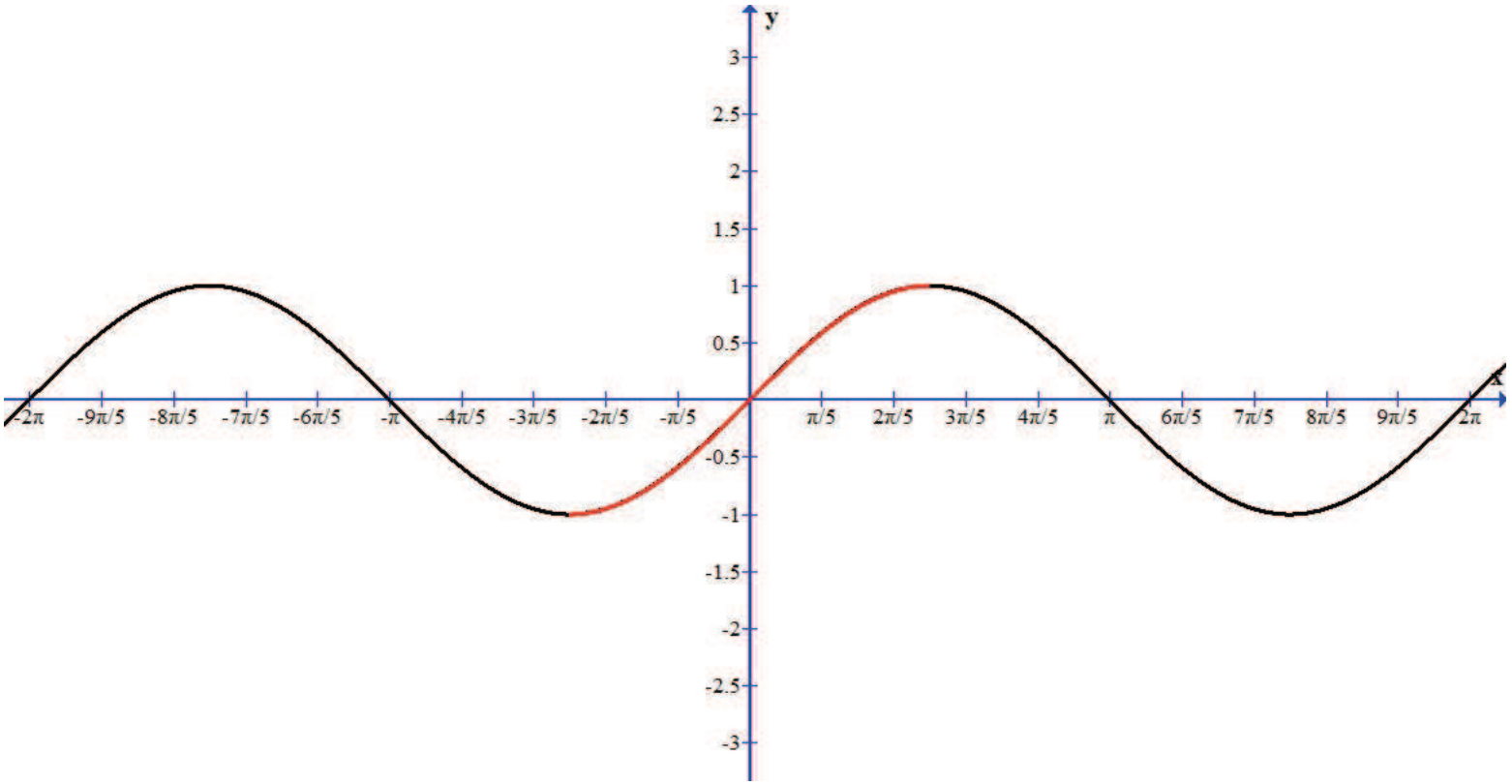
- $\sin 0 = 0\,,\ \cos 0 = 1\,,\ \sin \frac{\pi}{2} = 1\,,\ \cos \frac{\pi}{2} = 0$
- $\sin\,(x+y) = \sin x \cos y + \cos x \sin y$
- $\sin\,(x-y) = \sin x \cos y - \cos x \sin y$
- $\cos\,(x+y) = \cos x \cos y - \sin x \sin y$
- $\cos\,(x-y) = \cos x \cos y + \sin x \sin y$
- Питагорова теорема $\cos^2 x + \sin^2 x = 1$

Доказателство на картина



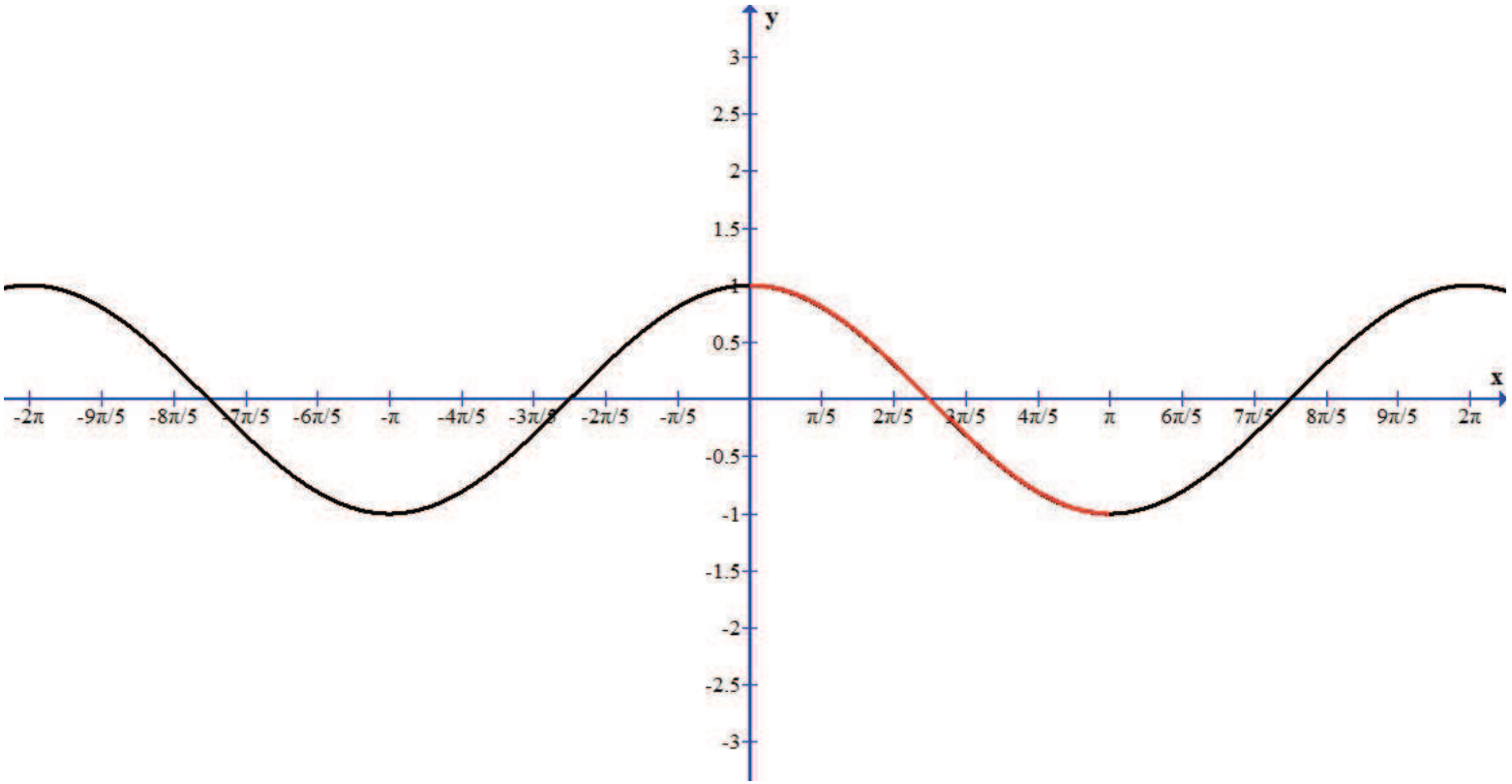
$\sin x$

– периодична с период 2π , нечетна, обратима в $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ (строго растяща)



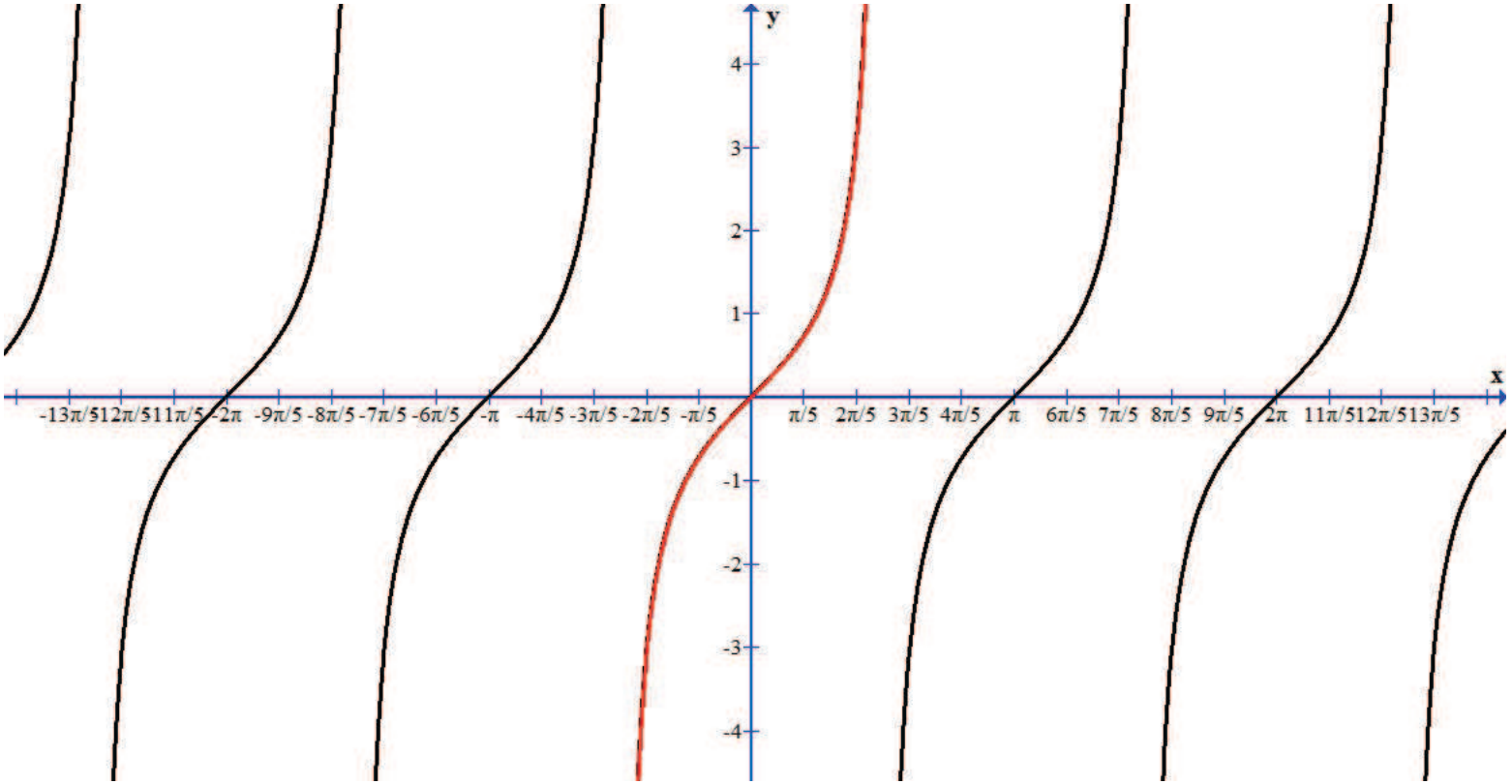
$\cos x$

– периодична с период 2π , четна, обратима в $[0\,,\,\pi]$ (строго намаляваща)



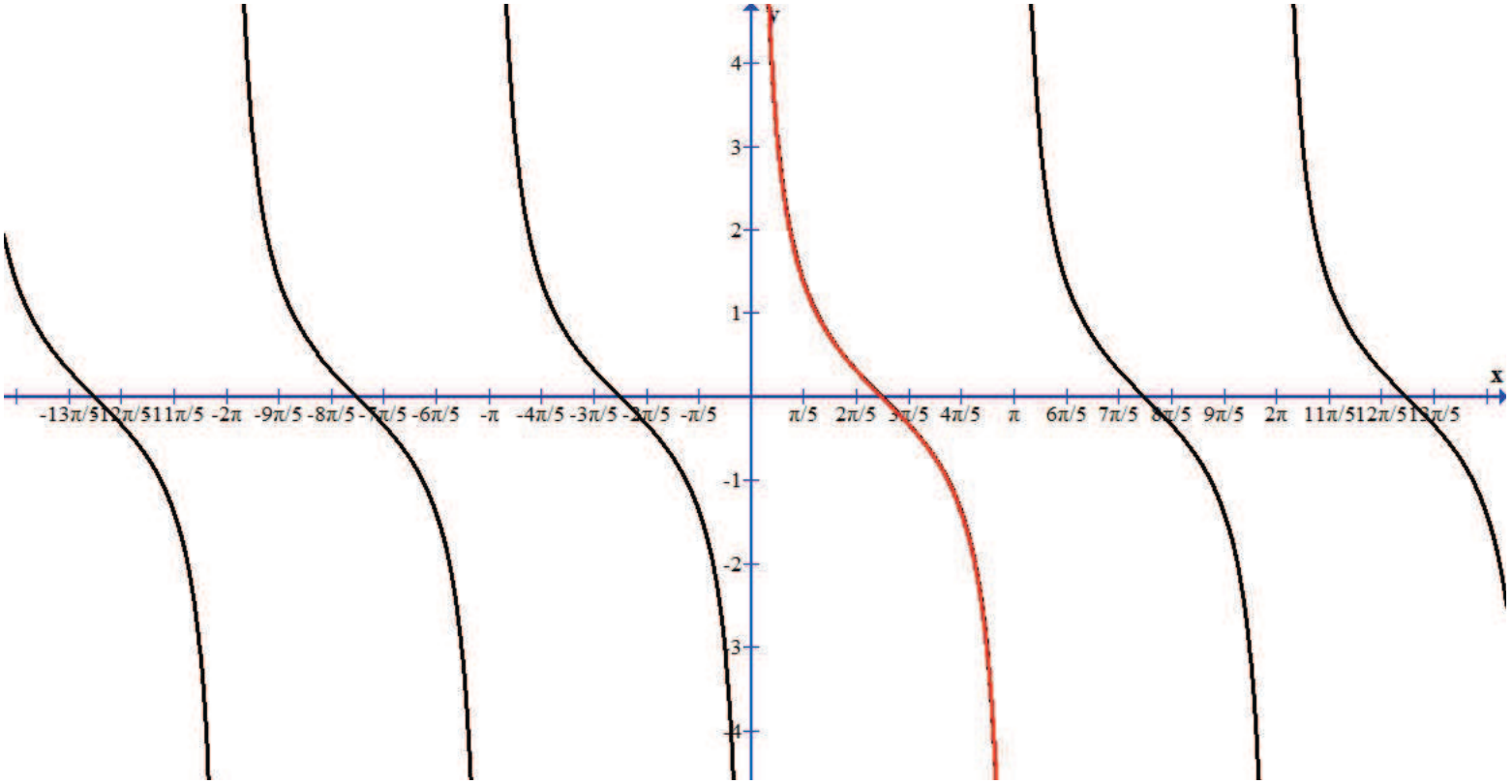
$\operatorname{tg} x$

$\operatorname{tg} x = \dfrac{\sin x}{\cos x}$ – периодична с период π, нечетна, обратима в $\left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right)$ (строго растяща)



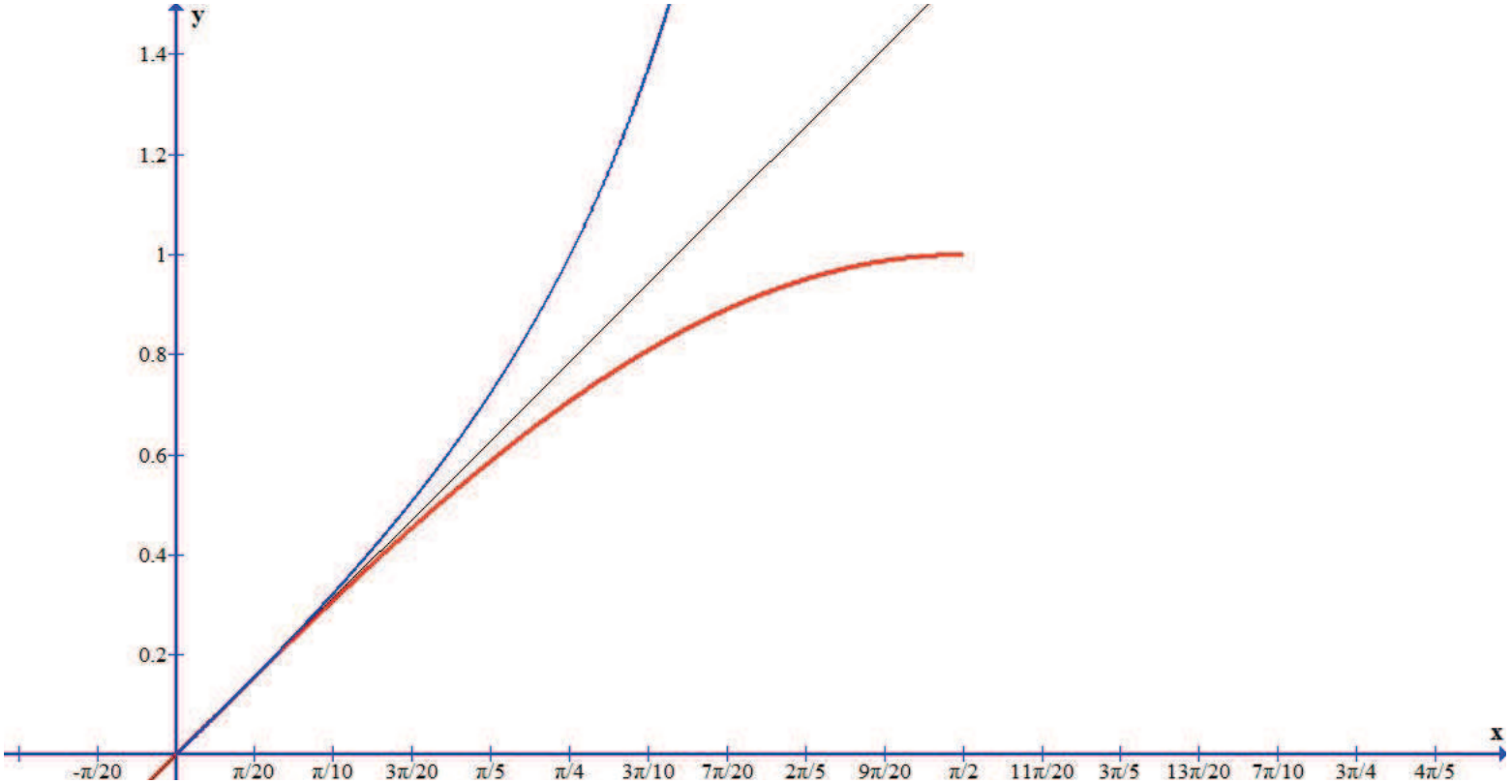
$\operatorname{ctg} x$

$\operatorname{ctg} x = \dfrac{\cos x}{\sin x}$ – периодична с период π, нечетна, обратима в $(0\,,\,\pi)$ (строго намаляваща)

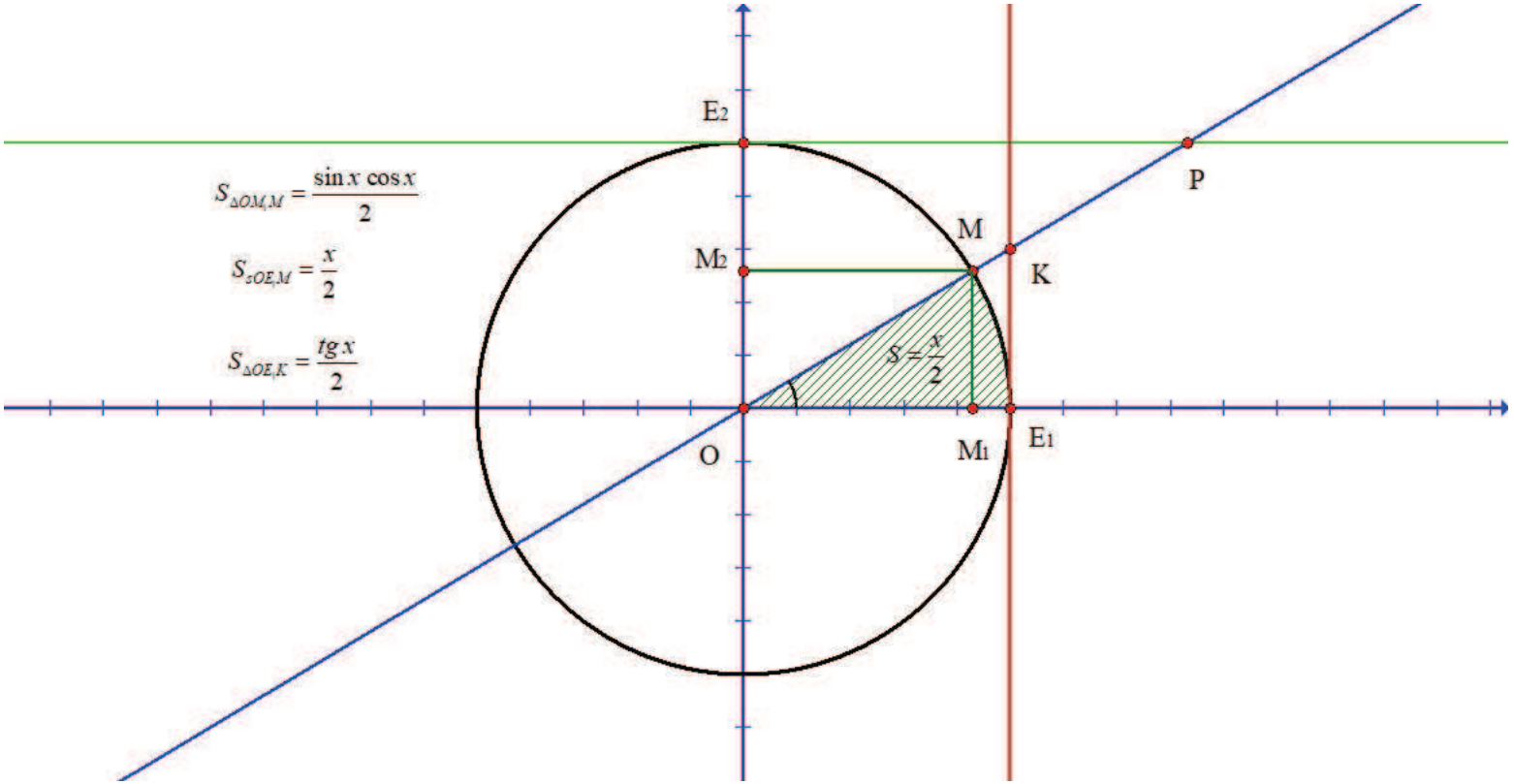


Основно неравенство

$0 < \sin x < x < \operatorname{tg} x$ – за всяко $x \in \left(0\,,\,\dfrac{\pi}{2}\right)$



Обосновка

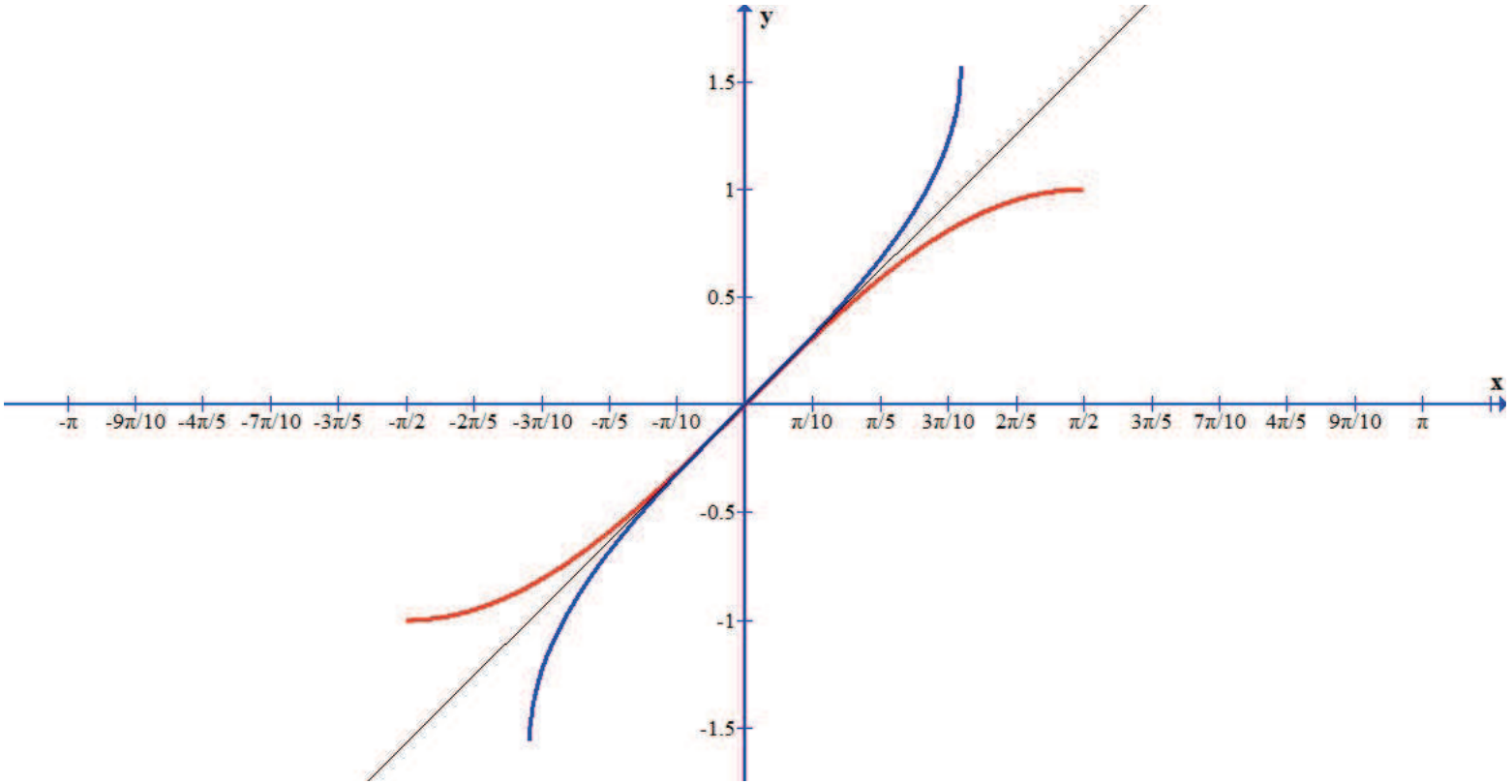

$S_{\Delta OM_1M} = \frac{\sin x \cos x}{2}$
$S_{sOE_1M} = \frac{x}{2}$
$S_{\Delta OE_1K} = \frac{tg\,x}{2}$
E2
P
M
M2
K
$S = \frac{x}{2}$
O
M1
E1

4.2 Обратни тригонометрични функции

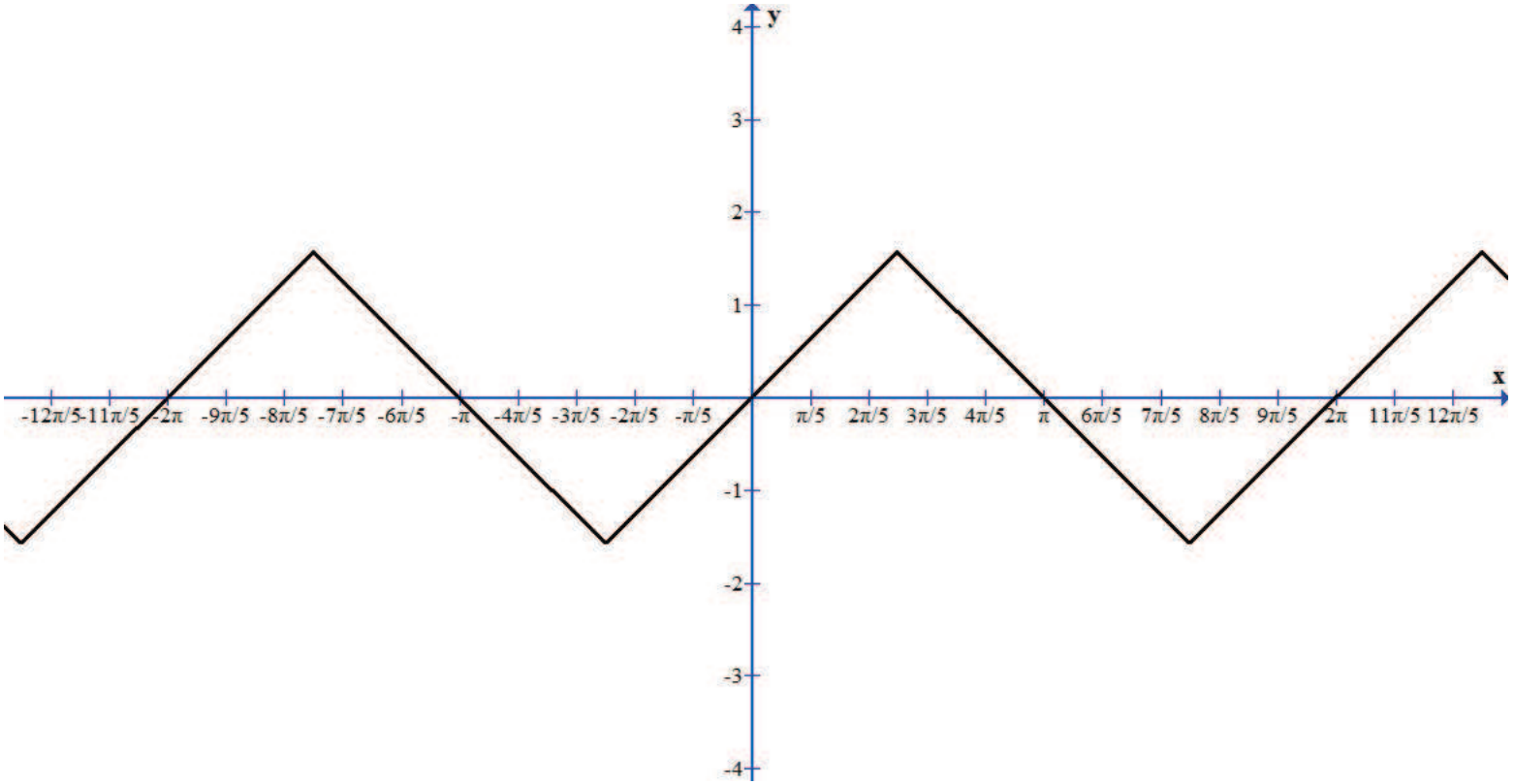
$\arcsin x$

– дефинирана в $[-1\,,\,1]$, стойности $\left[-\dfrac{\pi}{2}\,,\,\dfrac{\pi}{2}\right]$, строго растяща, нечетна

$\sin(\arcsin x) = x$ за $x \in [-1\,,\,1]\,,\quad \arcsin(\sin x) = x$ за $x \in \left[-\dfrac{\pi}{2}\,,\,\dfrac{\pi}{2}\right]$



Построяване на графиката на функцията $\arcsin(\sin x)$ – дефинирана навсякъде, периодична с период 2π, в $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ съвпада с x, а в $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$ – с $\pi - x$.

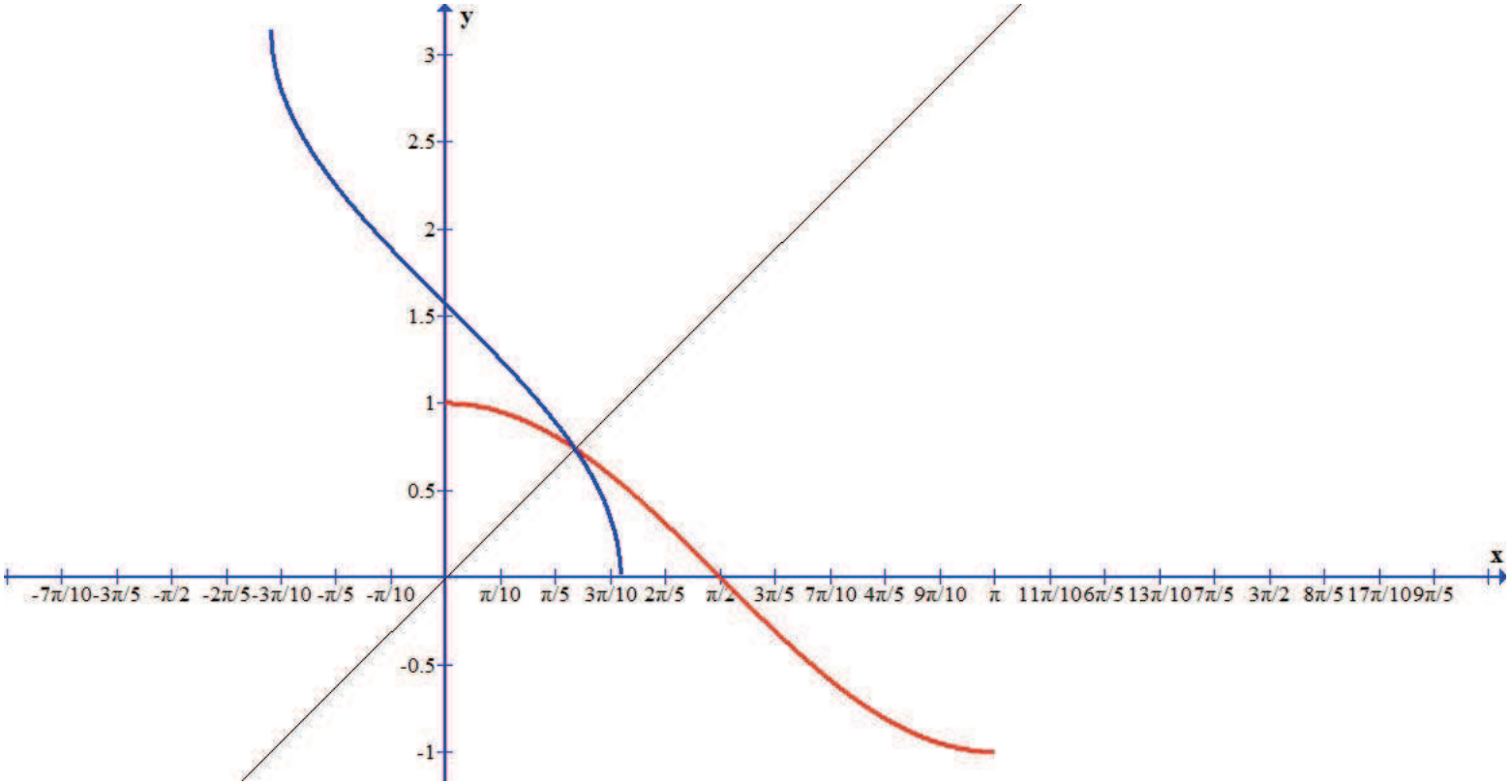


$\arccos x$

– дефинирана в $[-1\,,\,1]$, стойности $[0\,,\,\pi]$, строго намаляваща,

$\arccos(-x) = \pi - \arccos x\,, \quad \cos(\arccos x) = x$ за $x \in [-1\,,\,1]\,,$

$\arccos(\cos x) = x$ за $x \in [0\,,\,\pi]$

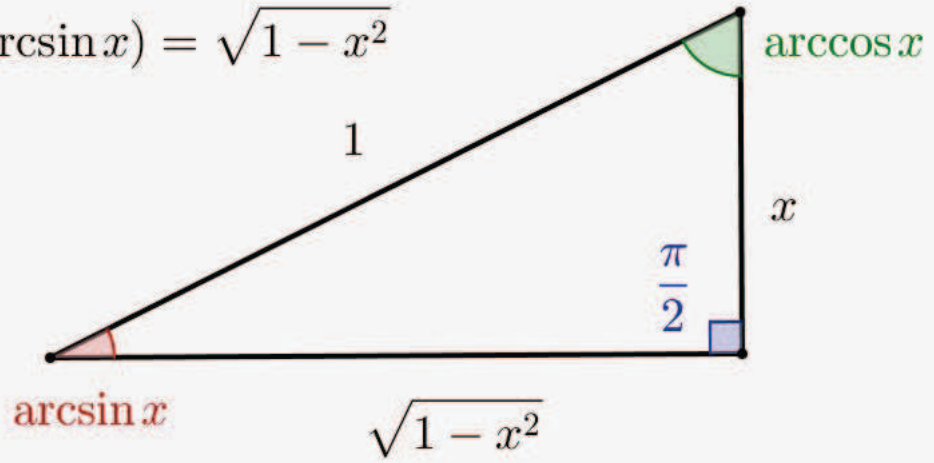


Връзки

$$\arcsin x + \arccos x = \frac{\pi}{2}$$

$$\sin(\arccos x) = \sqrt{1 - x^2}$$

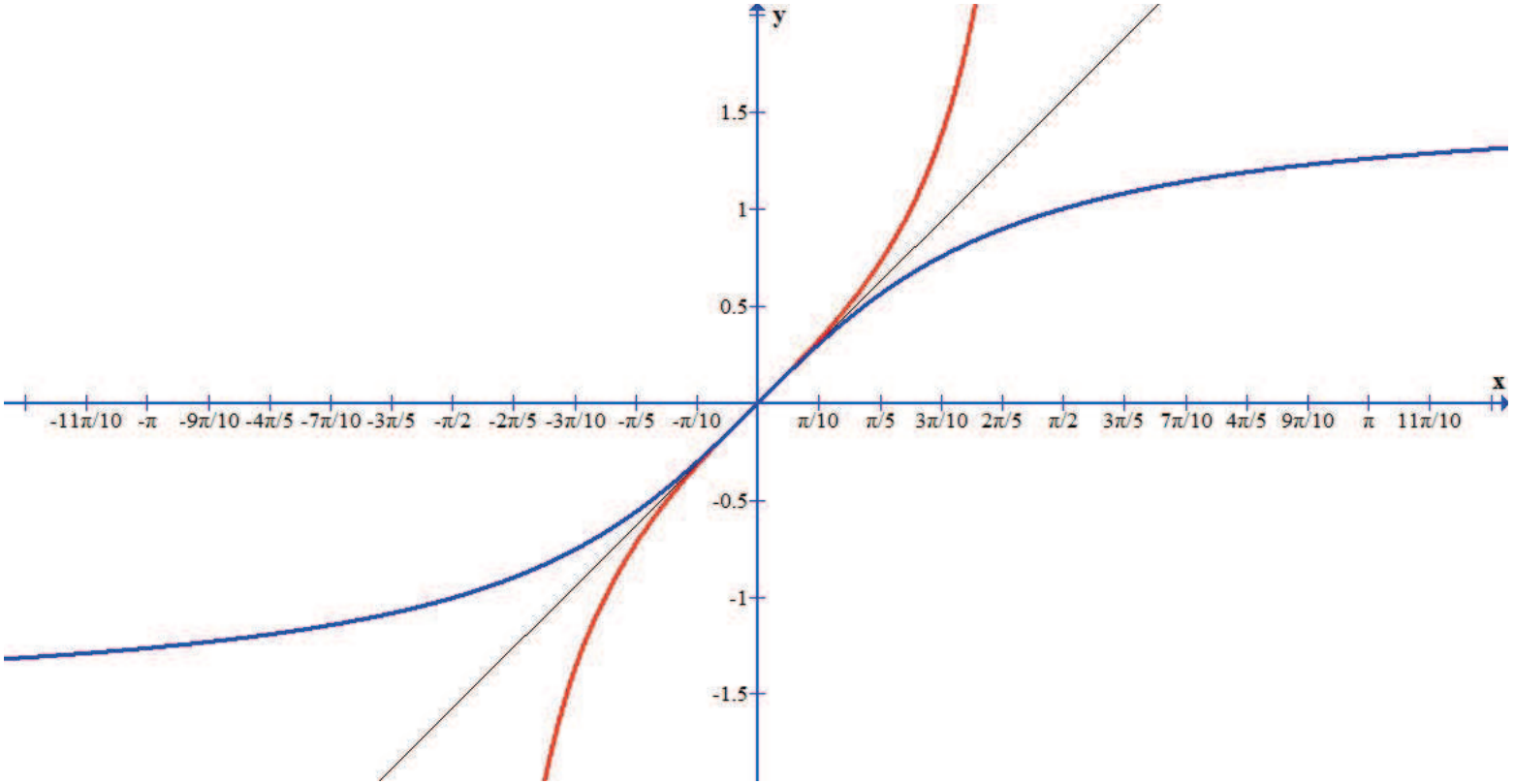
$$\cos(\arcsin x) = \sqrt{1 - x^2}$$



$\operatorname{arctg} x$

– дефинирана в $\mathbb{R}$, стойности $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, строго растяща, нечетна

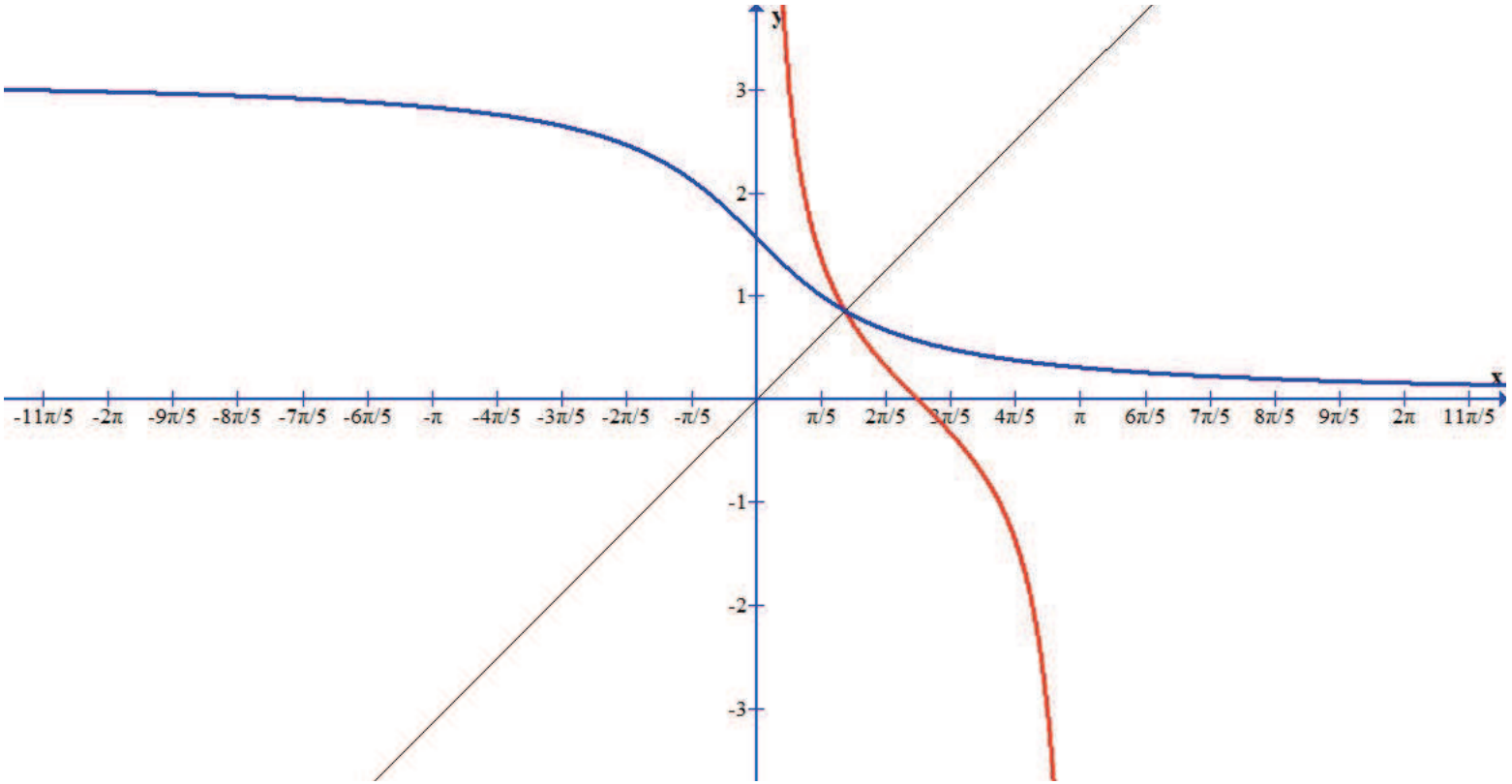
$\operatorname{tg}(\operatorname{arctg} x) = x$ за $x \in \mathbb{R}$, $\quad \operatorname{arctg}(\operatorname{tg} x) = x$ за $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$



$\operatorname{arcctg} x$

– дефинирана в $\mathbb{R}$, стойности $(0\,,\,\pi)$, строго намаляваща, $\operatorname{arcctg}(-x) = \pi - \operatorname{arcctg} x$

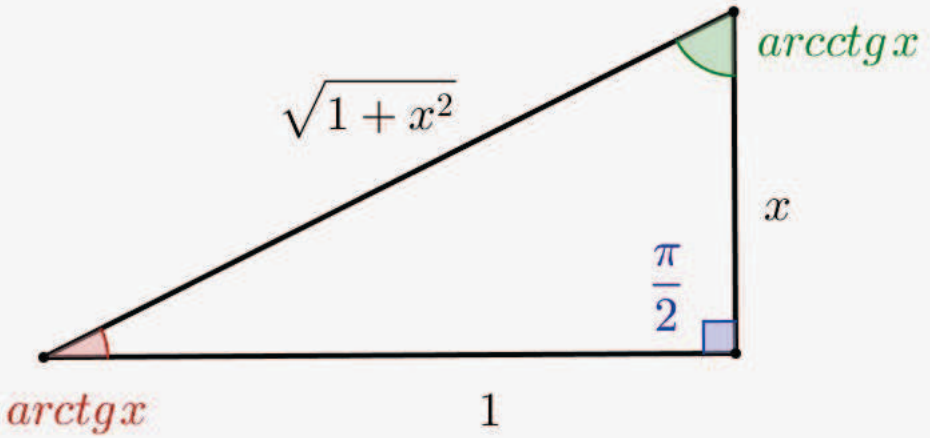
$\operatorname{ctg}(\operatorname{arcctg} x) = x$ за $x \in \mathbb{R}$, $\quad \operatorname{arcctg}(\operatorname{ctg} x) = x$ за $x \in (0\,,\,\pi)$



Връзки

$$arctgx + arcctgx = \frac{\pi}{2}$$

$$tg(arcctgx) = \frac{1}{x} \qquad cotg(arctgx) = \frac{1}{x}$$



$$sin\,(arctg\,x) = \frac{x}{\sqrt{1+x^2}} \qquad cos\,(arctg\,x) = \frac{1}{\sqrt{1+x^2}}$$

4.3 Доказателства и задачи

Две важни тъждества:

- $\sin\,(2\mathrm{arctg}\,x) = \dfrac{2x}{1+x^2}$ за $x \in \mathbb{R}$

  Доказателство: $\sin\,(2\mathrm{arctg}\,x) = 2\sin\,(\mathrm{arctg}\,x)\cos\,(\mathrm{arctg}\,x) = 2 \cdot \dfrac{x}{\sqrt{1+x^2}} \cdot \dfrac{1}{\sqrt{1+x^2}} = \dfrac{2x}{1+x^2}$

- $\cos\,(2\mathrm{arctg}\,x) = \dfrac{1-x^2}{1+x^2}$ за $x \in \mathbb{R}$

  Доказателство: $\cos\,(2\mathrm{arctg}\,x) = \cos^2(\mathrm{arctg}\,x) - \sin^2(\mathrm{arctg}\,x) =$

  $= \left(\dfrac{1}{\sqrt{1+x^2}}\right)^2 - \left(\dfrac{x}{\sqrt{1+x^2}}\right)^2 = \dfrac{1-x^2}{1+x^2}$

Някои от твърденията, цитирани преди:

- $\arcsin(\sin x) = \pi - x$ за $x \in \left[\dfrac{\pi}{2}, \dfrac{3\pi}{2}\right]$

Доказателство: $x \in \left[\frac{\pi}{2}\,,\,\frac{3\pi}{2}\right] \quad \Rightarrow \quad \alpha = (\pi - x) \in \left[-\frac{\pi}{2}\,,\,\frac{\pi}{2}\right]$,

$\beta = \arcsin(\sin x) \in \left[-\frac{\pi}{2}\,,\,\frac{\pi}{2}\right]$. $\quad \sin\alpha = \sin x = \sin\beta \;\Rightarrow\; \alpha = \beta$.

- $\arccos(-x) = \pi - \arccos x$

  Доказателство: $\arccos x \in [0,\,\pi] \;\Rightarrow\; \alpha = (\pi - \arccos x) \in [0,\,\pi]$, $\beta = \arccos(-x) \in [0,\,\pi]$.
  $\cos\alpha = -\cos x = \cos\beta \;\Rightarrow\; \alpha = \beta$.

- $\arcsin x + \arccos x = \frac{\pi}{2}$ за $x \in [-1\,,\,1]$

  Доказателство: $\arccos x \in [0,\,\pi] \;\Rightarrow\; \alpha = \left(\frac{\pi}{2} - \arccos x\right) \in \left[-\frac{\pi}{2}\,,\,\frac{\pi}{2}\right]$,

  $\beta = \arcsin x \in \left[-\frac{\pi}{2}\,,\,\frac{\pi}{2}\right]$. $\quad \sin\alpha = \cos\,\arccos x = x = \sin\beta \;\Rightarrow\; \alpha = \beta$.

- $\sin(\arccos x) = \sqrt{1 - x^2}$ за $x \in [-1\,,\,1]$

  Доказателство: $\arccos x \in [0,\,\pi] \;\Rightarrow\; \sin(\arccos x) \geq 0$.

  $\sin^2(\arccos x) = 1 - \cos^2(\arccos x) = 1 - x^2 \;\Rightarrow\; \sin(\arccos x) = \sqrt{1 - x^2}$.

- $\operatorname{tg}(\operatorname{arcctg} x) = \dfrac{1}{x}$ за $x \neq 0$

  Доказателство: $\operatorname{tg}(\operatorname{arcctg} x) = \dfrac{1}{\operatorname{ctg}(\operatorname{arcctg} x)} = \dfrac{1}{x}$.

- $\cos(\operatorname{arctg} x) = \dfrac{1}{\sqrt{1+x^2}}$ за $x \in \mathbb{R}$

  Доказателство: $\operatorname{arctg} x \in \left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right) \Rightarrow \cos(\operatorname{arctg} x) > 0$.

  $$\cos^2(\operatorname{arctg} x) = \frac{\cos^2(\operatorname{arctg} x)}{\sin^2(\operatorname{arctg} x) + \cos^2(\operatorname{arctg} x)} = \frac{1}{\operatorname{tg}^2(\operatorname{arctg} x) + 1} = \frac{1}{x^2+1}$$

  $$\Rightarrow \cos(\operatorname{arctg} x) = \frac{1}{\sqrt{1+x^2}} .$$

Задачи

- Да се пресметне: $\operatorname{arctg} 2 + \operatorname{arctg} 3$.

  Решение: $0 < \operatorname{arctg} 2 < \dfrac{\pi}{2}$, $0 < \operatorname{arctg} 3 < \dfrac{\pi}{2} \Rightarrow 0 < \operatorname{arctg} 2 + \operatorname{arctg} 3 < \pi$.

$\cos\,(\operatorname{arctg}2+\operatorname{arctg}3)=\cos\,(\operatorname{arctg}2)\,.\cos\,(\operatorname{arctg}3)-\sin\,(\operatorname{arctg}2)\sin\,(\operatorname{arctg}3)=$

$=\dfrac{1}{\sqrt{5}}\cdot\dfrac{1}{\sqrt{10}}-\dfrac{2}{\sqrt{5}}\cdot\dfrac{3}{\sqrt{10}}=-\dfrac{1}{\sqrt{2}}\quad\Rightarrow\quad\operatorname{arctg}2+\operatorname{arctg}3=\dfrac{3\pi}{4}\ .$

- Да се докаже, че $\operatorname{arctg}x+\operatorname{arctg}y=\operatorname{arctg}\dfrac{x+y}{1-xy}$ за $xy<1$.

Решение: $-\pi<\operatorname{arctg}x+\operatorname{arctg}y<\pi$, $\cos\,(\operatorname{arctg}x+\operatorname{arctg}y)=$

$=\cos\,(\operatorname{arctg}x)\,.\cos\,(\operatorname{arctg}y)-\sin\,(\operatorname{arctg}x)\sin\,(\operatorname{arctg}y)=$

$=\dfrac{1}{\sqrt{1+x^2}}\cdot\dfrac{1}{\sqrt{1+y^2}}-\dfrac{x}{\sqrt{1+x^2}}\cdot\dfrac{y}{\sqrt{1+y^2}}=\dfrac{1-xy}{\sqrt{1+x^2}\,.\sqrt{1+y^2}}>0$

$\Rightarrow\quad\alpha=(\operatorname{arctg}x+\operatorname{arctg}y)\in\left(-\dfrac{\pi}{2}\,,\,\dfrac{\pi}{2}\right).\qquad\beta=\operatorname{arctg}\dfrac{x+y}{1-xy}\in\left(-\dfrac{\pi}{2}\,,\,\dfrac{\pi}{2}\right),$

$\operatorname{tg}\alpha=\dfrac{\operatorname{tg}\,(\operatorname{arctg}x)+\operatorname{tg}\,(\operatorname{arctg}y)}{1-\operatorname{tg}\,(\operatorname{arctg}x)\,.\operatorname{tg}\,(\operatorname{arctg}y)}=\dfrac{x+y}{1-xy}=\operatorname{tg}\beta\quad\Rightarrow\quad\alpha=\beta\ .$